Natural Expression of a Machine Learning Model's Uncertainty Through Verbal and Non-Verbal Behavior of Intelligent Virtual Agents

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Figure 1: Illustration of the natural expression of uncertainty through intelligent virtual agents: *(top left)* A user question with *(bottom left)* an incorrect answer as generated by Google Gemini (Pittsburgh would be the correct answer). The chatbot provides no indication of the ML model's confidence. *(right)* Our ML model that accompanies (potentially incorrect) answers with lexical and visual uncertainty cues displayed via an agent. Agent model by Reallusion Inc.

ABSTRACT

Uncertainty cues are inherent in natural human interaction, as they signal to communication partners how much they can rely on conveyed information. Humans subconsciously provide such signals both verbally (e.g., through expressions such as "maybe" or "I think") and non-verbally (e.g., by diverting their gaze). In contrast, artificial intelligence (AI)-based services and machine learning (ML) models such as ChatGPT usually do not disclose the reliability of answers to their users.

In this paper, we explore the potential of combining ML models as powerful information sources with human means of expressing uncertainty to contextualize the information. We present a comprehensive pipeline that comprises (1) the human-centered collection of (non-)verbal uncertainty cues, (2) the transfer of cues to virtual agent videos, (3) the annotation of videos for perceived uncertainty, and (4) the subsequent training of a custom ML model that can generate uncertainty cues in virtual agent behavior. In a final step (5), the trained ML model is evaluated in terms of both fidelity and generalizability of the generated (non-)verbal uncertainty behavior.

CCS CONCEPTS

• **Human-centered computing** → **Natural language interfaces**; *Virtual reality*.

KEYWORDS

virtual agents, transparency in machine learning, human communication

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1 INTRODUCTION

As technology advances, our primary sources of information continue to evolve; from oral speech to written artifacts and print to electronic media [58]. Driven by the popularization of large language models (LLMs) such as OpenAI's ChatGPT or Google's Gemini, we are currently observing another paradigm shift - from human-written texts to texts generated by machine learning (ML) models [14]. In a study conducted in March 2023 with 60 students [37], 70% of participants stated that they use ChatGPT as an information retrieval tool, with 58% of all participants rating its effectiveness as better than traditional search engines such as Google. One aspect that is often mentioned alongside the opportunities offered by LLMs to support information retrieval is the risk of spreading misinformation, especially for users that are not aware of the underlying technological principles [54].

While misinformation is not a problem exclusive to ML, other information sources come with more advanced strategies to mitigate the risk of passing on incorrect knowledge. For example, in human-to-human communication, when a person answers a (factual, mathematical, reasoning-based, ...) question, they intentionally or unintentionally accompany it with verbal or non-verbal indications of how certain they are about their response [23]. Such uncertainty cues are often encoded in linguistics (e.g., uttering words like "probably" or "maybe", phrases like "I guess", filler words, hedges), prosody (e.g., rising intonation, low speech rate), or movements (e.g., raising eyebrows, diverting gaze) [23].

The concepts of uncertainty and probability are also inherent in various forms of ML models [21]. For example, during image classification, a commonly used output are probability distributions over the image classes [10, 18, 27]. Similarly, LLMs use the probability distribution over all possible tokens to predict the next token during text generation [63]. Numerous research projects are concerned with how such probabilities can be directly or indirectly converted into an interpretable representation of the uncertainty of the ML model (cf. Section 2.1). Nevertheless, when using current AI-based services, such as Amazon Alexa or Apple's Siri, these values are often not made available to the end user, making it challenging to quantify how reliable a model's prediction is [4, 41]. Providing additional cues about the certainty of responses could increase the trustworthiness, transparency as well as explainability of AI/ML-generated content [8, 12]. This potential is even more evident for embodied intelligent virtual agents, when human-computer interaction is based on natural means of communication, such as speech, voice, gestures, or facial expressions.

In this paper, we present the first extensive investigation on whether the human concept of (non-)verbal expression of uncertainty could be automatically transferred to a virtually embodied AI system (hereinafter referred to as virtual agent). In particular, we consider the following research questions:

- Which human uncertainty cues are suitable for expressing the uncertainty of an AI system?
- How nuanced can the uncertainty of an AI system be expressed by an embodied virtual agent through the combination of verbal and non-verbal cues?
- Do multimodal uncertainty cues increase the accuracy of the uncertainty conveyed by the virtual agent?
- How universal is the interpretation of uncertainty cues across virtual agents of different gender, age, or ethnicity?

To address these research questions, we developed a pipeline with a sequence of several user-centered and technical stages. First, we collected uncertainty cues in verbal behavior (Section 3.1) and non-verbal behavior (Section 3.2) of users based on trivia quizzes. We transferred these cues to virtual agents (Section 3.3), which were then shown to users, who annotated the uncertainty they perceived (Section 3.4). The combination of uncertainty cues and annotated uncertainty served as a training basis for a custom ML model, which is designed to generate verbal and non-verbal uncertain behavior displayed via a virtual agent (Section 3.5). In a perceptual evaluation (Section 3.6), we were able to show that despite the amount of available training data being limited compared to the resources of large AI companies, it was possible to train an ML model that can express a range of uncertainty levels in the form of (non-)verbal agent behavior. Based on various data analyses, we discuss how this proof of concept can be extended to practical models in the future and which aspects need to be considered for generalization to different cultures, languages or agent types (Section 4). Throughout the process, we adhere to standards for the definition of virtual agents so that our openly available model[1] can also be used by other researchers for investigations with their own virtual agents.

2 RELATED WORK

In the following, we provide an overview of previous research on uncertainty in different contexts: machine learning, classical human-human interaction, and virtual agents.

2.1 Uncertainty in Machine Learning

The quantification of uncertainty or confidence of ML and deep learning (DL) models is a much-studied subject in research [1, 21], especially in application contexts where interpretability of the output is of major importance. These include, for example, healthcare [13], visualization [41], and facial expression recognition [45, 84]. Confidence in ML is directly linked to the concept of truthfulness, i.e. whether the model provides (factually) accurate outputs [36]. Uncertainty can occur at different stages of the ML pipeline, during data collection, model building, training, inference, or the predictions' uncertainty mode [21].

For a (calibrated) classification model, the output would be a probability distribution over all class labels, with the confidence quantifying how certain the model was with this output [21]. In contrast, for a regression problem, the model would return a probability distribution with mean and standard deviation [21], which is a direct measure of uncertainty. For the construction of probabilistic ML models, multiple paradigms have been proposed, such as Bayesian Neural Networks that directly quantify the uncertainty of the output.

[1] Our open-source ML model for generating uncertain agent behavior is available on: https://github.com/uhhhci/deep-learning-for-uncertain-agent-behavior.

LLMs also involve probability distributions, but their interpretation is less straightforward than for the previously mentioned models. Models such as ChatGPT output a sequence of tokens along with information about how likely each token is at each position in the overall response [48]. These probability distributions therefore relate to the lexical level and do not indicate whether the overall response is correct on a semantic level [44].

Kuhn et al. [44] link the two concepts of lexical and semantic uncertainty by identifying clusters of responses that have the same meaning and calculating the semantic entropy for these clusters based on the token probabilities of the cluster members. A similar approach, though not requiring token-level probabilities, was presented by Lin et al. [49]. The authors generate multiple output samples for each input and then calculate a confidence value based on the pairwise similarity scores between these samples.

Another line of research to quantify the confidence of LLMs is self-evaluation, i.e. prompting the model to estimate the probability that its own output is correct. Kadavath et al. [36] have demonstrated that LLMs perform well when they are asked to self-assess whether their generated responses are true or false. Lin et al. [48] took this a step further by examining verbalized probability, i.e. asking LLMs to express their confidence in a particular response in a human-like manner. By fine-tuning GPT-3, Lin et al. [48] were able to achieve a reasonable calibration for arithmetic tasks, with the model producing confidence values in percentages.

While the aforementioned approaches still have certain limitations, such as a lack of generalizability to tasks other than those evaluated, all have been published within the last two years, highlighting current efforts to increase the truthfulness of LLMs. These are important because LLMs still produce factually inaccurate responses, as exemplified in Figure 1, and extrinsic instructions, such as Google Gemini's disclaimer that answers should be double-checked, may be forgotten or overlooked by users [53]. Even if such factual errors are expected to become less frequent as LLMs improve, Mielke et al. [53] argue that the ability to express uncertainty will always remain relevant, as there will be questions with unknown answers or with answers that depend on a context that the agent cannot know. In our project, we therefore directly address the natural representation of uncertainty for the users of ML models, abstracting from the specific implementation of uncertainty quantification and referring to the related literature instead.

2.2 Uncertainty Visualization

Even for ML architectures that already allow a proper quantification of their uncertainty, its communication to users proves challenging [8, 59]. This is particularly the case as end users of current ML models are oftentimes non-experts, both in everyday contexts such as weather forecasting or transit arrival times predictions, and in high-stake contexts such as medical diagnosis, elections, or climate change [8, 29]. Even without the technical literacy, users need to properly understand the uncertainty involved in provided information to derive how much they should trust model predictions and consequently to make informed, rational decisions.

According to Van Der Bles et al. [75], uncertainty about a fact, number or scientific hypothesis can be expressed using a numerical, verbal, and/or visual format. Which format to use is not only a design choice, as it is both determinative of the carried level of information and is directly influencing the interpretation of the audience.

Numerical representations of uncertainties, such as (ranges of) probabilities (e.g., "10%" or confidence intervals) and frequencies (e.g., "1 in 10") generally hold a high precision [75], but not necessarily a high accuracy [8]. In a study by Galesic and Garcia-Retamero [20], even basic statistical indicators were shown to be subject of misunderstandings. Approximately 20% of study participants could not correctly answer whether 1%, 5%, or 10% represents the largest risk of getting a disease; for frequencies (i.e., 1 in 10, 1 in 100, or 1 in 1000) the percentage of incorrect answers even increased to almost 30%. A similar study aiming at mostly well-educated participants still yielded error rates of 16% and 22% for probabilities and frequencies, respectively [50]. In addition, a variety of cognitive biases affect the interpretation of uncertainty estimates, such as tendencies to underestimate high probabilities while overestimating low probabilities, as well as preferences towards positive framing [8].

Verbal probability qualifiers like "most", "likely", or a "majority" are oftentimes preferred over numerical expressions by both experts and non-experts [79], although generally yielding a lower precision [75, 77]. Studies revealed considerable differences in the interpretation of such expressions, not only between participants [77] but also for the same person in different contexts [11]. For example, when presented with different medical events (e.g., developing a rare severe migraine or a common mild influenza), the base rate and perceived severity/consequentiality of the condition were significantly affecting probability estimates for expressions such as "likely" [79]. When asked about the range of probabilities that could be associated with the expression "likely", participants of a study by Wallsten et al. [77] indicated median values for the lower and upper limit of 0.59 and 0.9, respectively.

For the third format of uncertainty expressions, i.e. visual representations, Padilla et al. [59] distinguish between two subcategories. Firstly, graphical annotations represent properties of a distribution, with typical visualization elements being error bars or boxplots. Secondly, uncertainty can be encoded in a visual attribute, such as color, position, or transparency. Both approaches can also be combined to form a hybrid visual representation. As with numerical values, it has been shown that uncertainty visualizations are subject to several interpretation biases [35]. One challenge, especially for non-expert users, is to even recognize that a visualization expresses uncertainty. An example of this so-called deterministic construal error [35] concerns the visualization of the possible course of a hurricane by means of an uncertainty cone, which is widely misinterpreted as the diameter of the storm, increasing over time [65]. While there are also specific visualization methods for communicating errors of ML models to experts, such as ROC and Precision-Recall curves, these were shown to be difficult to understand for non-expert users [5, 6].

Overall, Van Der Bles et al. [75] argue that there are nine different levels for communicating uncertainty with decreasing precision, including (i) a full explicit probability distribution, (v) a

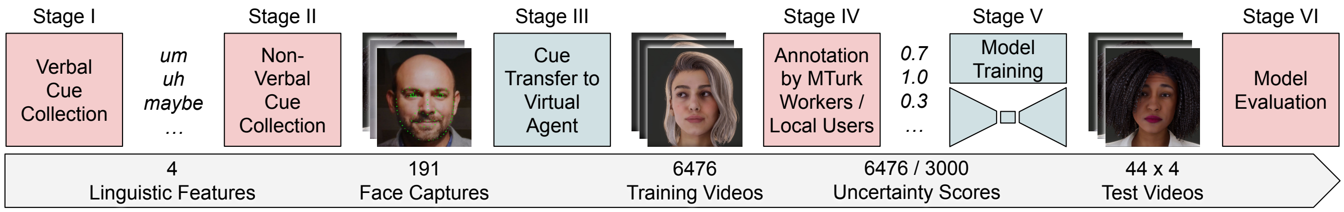

Figure 2: Project pipeline with the red stages involving human participation. Agent models by Reallusion Inc.

qualifying verbal statement, and (vii) informally mentioning the existence of uncertainty. Results of a study by Greis et al. [25] suggest that presenting a higher degree of uncertainty information to non-experts does not necessarily result in a higher perceived support for decision-making. A conclusion especially relevant to our own research is that high-precision communication methods can help experts understand the full extent of uncertainty in ML models, while low-precision methods may be sufficient for non-experts with potentially limited numeracy skills [8].

2.3 Uncertainty in Human Interactions

The results presented above suggest advantages of using intuitively interpretable uncertainty cues, especially for non-experts. Suitable cues could be those known from human communication, as people respond to them subconsciously [72]. Uncertainty, or the lack of confidence, has already been studied in classical psychology and linguistics for several decades. Much of the more recent work in this area is based on a study by Smith and Clark [68], which analyzed the linguistic structure of study participants' answers to general knowledge questions. Smith and Clark [68] found evidence that question answering does not only consist of a memory search for an adequate answer and responding, but has an intermediate stage of monitoring one's own metamemory. In this second stage, the respondent is judging how confident they are that the potential answer retrieved from their memory is correct and whether they should proceed with the memory search. Based on this judgement, certain cues can signal the respondent's current metacognitive state and their level of uncertainty to the listeners. These include linguistic hedges, such as "I guess" or "perhaps," and a rising, question-like intonation [68]. In addition, the prolonged delay required for a longer memory search can be compensated for by interjections, sometimes also referred to as fillers (e.g., *uh*, *um*, *hm*, *mm*), self-talk and explicit commentary [68].

In subsequent studies (partially) replicating the original work by Smith and Clark [68], further non-lexical uncertainty cues were identified. Prosodic features, especially in the temporal domain (e.g., silence and speaking duration), were found to be strongly associated with the perceived level of certainty [61]. Visual features, including the presence of an eyebrow movement, gaze movement, and a "funny face" (i.e., a marked facial expression, which diverts from a neutral expression), also correlate with a lower feeling of knowing [71]. Swerts and Krahmer [71] further found that a combination of both auditory and visual cues improve the accuracy of perceived uncertainty ratings.

2.4 Uncertainty in Virtual Agents

According to the computers-are-social-actors (CASA) paradigm, humans show comparable social behavior (including similar expectations, assumptions and attributions) towards a computer (agent) as towards other human interaction partners [55]. In line with this paradigm, previous studies suggest that various findings from human communication translate well to embodied virtual agents [78].

Several projects have thus attempted to utilize the previous findings from human communication to create more natural human-machine interactions, however, not necessarily with a focus on simulating uncertainty. Google Duplex, a voice agent that performs phone calls for tasks such as scheduling an appointment, incorporates speech disfluencies (e.g., *hmm* and *uh*) and varying levels of latency to generate more natural-sounding speech [46]. Similarly, Andersson et al. [3] collected data on the prediction of fillers such as *um*, *uh*, and *you know* in spontaneous speech and were able to show in a user study that the speech synthesized with their approach was perceived as closer to everyday conversations than read aloud speech, with no loss of naturalness. Other models for speech synthesis have focused on the insertion of silent pauses as a means of hesitation [7], as well as on the modulation of prosodic features such as speech rate [40].

In a series of user studies, Wollermann et al. [83] directly investigated whether variations in intonation, filler words and delay in synthetic speech contribute to the perception of uncertainty in a way comparable to natural human-human conversations. Their results suggest that the tested cues are not only effective, but that there is an additive effect on uncertainty perception when multiple cues are integrated into one synthetic utterance. However, no correlation was found between perceived uncertainty and naturalness.

Although visual cues have been found to increase the accuracy of uncertainty perception (see Section 2.3), studies on agents visualized with a virtual body are still scarce. Previous studies on human-agent interaction without a focus on uncertainty have shown beneficial effects of embodiment compared to voice-only agents, for example a lower cognitive load for the user [38], as well as a higher perceived agent credibility [66], and higher trust in the agent [80]. Through non-verbal feedback, embodied agents can provide additional social cues, for example, that they are thinking [15] or looking for an appropriate response to a user's request [67].

In the scope of a larger project aimed at developing a real-time facial animation system called RUTH, Stone and Oh [70] explored the possibilities and challenges of transferring uncertain verbal and non-verbal behavior (facial expressions, gaze and head pose) to a simplified agent head model. In their exploratory experiments, they found that the range of motions of RUTH is insufficient to represent the complex facial expressions that their study participants displayed during uncertain responses. They also found that the lack of skin features like wrinkles is a limiting factor that makes it difficult for users to distinguish uncertainty from other emotions such as anger, sadness, surprise, or fear.

In a related paper, Oh and Stone [57] present a study in which they manually transferred 10 speaker videos with different levels of uncertainty to RUTH. Despite the highly simplified and stylized appearance and motion of RUTH, the authors found high correlations between the perceived uncertainty in RUTH's behavior and the self-reported uncertainty of the original speaker.

Marsi and Van Rooden [52] came to similarly positive results when integrating audiovisual uncertainty cues into a Q&A demonstrator, also based on RUTH. Their preliminary results indicate that study participants were able to correctly discern certain and uncertain facial expressions, although no intermediate states between these two extremes were examined.

2.5 Contextualization of Own Research

In summary, an ML model's uncertainty can be quantified (see Sec. 2.1) and visualized (see Sec. 2.2) to increase its interpretability and truthfulness. High-precision visualizations are cognitively demanding and widely misinterpreted by non-expert users (see Sec. 2.2) while humans are trained to intuitively interpret communication cues of interlocutors, with multimodality increasing the accuracy of uncertainty estimations (see Sec. 2.3). Based on the CASA principle, virtual agents could utilize such human communication cues, with previous studies showing that speech and (manually animated) facial expressions can elicit different perceptions of uncertainty when modulated separately (see Sec. 2.4).

Based on these previous inter-disciplinary findings, our goal was to jointly simulate verbal and non-verbal cues and to investigate whether the generated multimodal feedback can be interpreted consistently. Since it is not feasible to train an existing ML model (e.g. LLaMA-2) to generate textual and visual cues (cf. Sec. 3.5.5), one of the main challenges was to construct and train a custom ML model. Therefore, in contrast to previous projects, we investigate the feasibility of predicting (1) multimodal uncertainty behavior (2) of a high-fidelity embodied virtual agent (3) using a custom-trained ML model.

3 PROJECT PIPELINE

With the overall goal of training an ML model that can generate varying verbal and non-verbal behavior of a virtual agent based on an input uncertainty value, we went through six stages, four of which involved collecting empirical data from users (see Figure 2). We started by recording multimodal training data in a series of human-based studies. All data collection steps received ethical approval from the local ethics commission.

3.1 Stage I: Collection of Verbal Cues

In a first step, we focused on identifying uncertainty cues that occur most frequently in a question answering scenario and at the same time are suitable to be transferred to a virtual agent. It is already known from the literature that such cues are diverse in their form (e.g., lexical, prosodic, visual) and that even within one modality a large number of specific types (e.g., filler words, hedges, self-talk) and type levels (e.g., *um*, *uh*, *mm*, *hm*) exist. Since the study was conducted with native German speakers and the objective was to capture natural uncertainty behavior, the first two parts of data collection were conducted in German (possible resulting confounds are discussed in Section 4).

Pipeline Stage 1: Collecting Verbal Cues

Input: None

Output: Four categories of verbal uncertainty cues, each with multiple levels (see Table 1)

Task:

1. Collect videos of users answering trivia questions.
2. Transcribe videos.
3. Discard videos with non-answers.
4. Analyze transcripts using hybrid coding.
5. Select suitable verbal uncertainty cues based on criteria such as high frequency and generalizability.

3.1.1 *Materials.*

Following the example of related experiments on uncertainty from classical linguistic research (see Section 2.3), we conducted a trivia quiz with participants. It is based on a corpus of 299 general knowledge questions, each of which is accompanied by an average self-reported confidence score [73]. For our quiz, we selected 30 questions that cover the entire confidence range. We also tried to minimize the frequency of non-answers from our study participants by selecting questions that allowed for guessing (e.g., asking for a country or planet rather than the name of a specific series character).

For conducting the quiz, we developed a customized web application that allows researchers to collect video data from users who spontaneously react to pre-recorded stimuli (in this case the trivia questions). Since the further processing steps of the recorded video data prohibit full anonymization, we placed particular emphasis on aspects such as secure data storage and transmission as well as user control over the data that is collected.

All selected trivia questions were translated into German using DeepL [16] and recorded by a female speaker. Based on previous related studies in which the experimenter asking the questions did not face the participants to avoid visual cues [9, 71], we did not record a video of the experimenter, but only showed a question mark alongside the audio track.

3.1.2 *Methods.*

Prior to the study, participants had to read and sign a consent form, in which particular emphasis was placed on the purpose, protection and confidentiality of the identifiable (video/audio) data. The study itself was conducted using a laptop or computer in a quiet environment selected by the participant. The experimenter briefly introduced the study objective and the recording app either on site or via a video conferencing system. The experimenter then left the room or, if the evaluation took place online, muted themselves and switched off the camera. During the trivia quiz, the entire study flow was controlled by the app itself. The response time for each question was limited to 30 seconds. Participants could choose whether they wanted to delete a video immediately after recording or at the end of the session, where they also had to actively confirm that undeleted video responses would be uploaded to the server. However, we emphasized that all responses were helpful for our research. The quiz took approximately 10 to 20 minutes per participant.

3.1.3 Participants. 18 participants took part in the first study. 12 of them were enrolled as (doctoral) students at the time of data collection. The age distribution was 18-24 years ($N = 8$), 25-34 years ($N = 8$), and 45-54 years ($N = 2$).

3.1.4 Analysis and Results. A total of 516 participant responses were collected. These were first transcribed using the offline Whisper large-v2 model [62] and then manually corrected to capture all subtleties such as intonation and fillers. We then followed a content analysis approach with hybrid (deductive and inductive) coding of all transcripts by the first author. The initial codes were based on literature in the field of linguistics, including filler words / interjections [68], various forms of hedges (speculative verbs [22], modal auxiliary verbs [17, 22], epistemic adverbs [17, 22]) and intonation [68]. During the first coding pass, all transcripts were marked as answers or non-answers (i.e., no response was given to the trivia question) and each answer ($N = 396$) was coded. Additional inductive codes were developed in this first pass: Hedge phrases, partial or complete repetition of the question, repetition of the answer, commentary/self-talk, and presentation of alternatives. With the exception of the first, all of these categories were coded with binary decisions. The codes *speculative verbs* and *epistemic adverbs* were also each divided into phrases that preceded or followed the response. In a second pass, all answers (as opposed to non-answers) were (re)coded with the final set of codes. Tables 3 and 4 in the supplementary material contain a list of all codes, examples in German and English, and frequencies.

From the coded responses, we selected features based on (1) high frequency, (2) generalizability to other scenarios besides quizzes, (3) transferability to virtual agents, and (4) potential impact on user experience in human-agent interactions. For example, "I log in" is a typical response for quizzes, but unusual for everyday conversations (cf. (2)). We further found that many responses were lengthy and contained a lot of repetition and comments on the participant's associations with a question (e.g., past vacations, related jokes, ...). Since we suspect that the user experience could be adversely affected by agents that extremely prolong the actual answer and present invented background stories, we also excluded these (see (4)). Table 1 shows the features we selected for our model.

To create a model that is more useful to the international research community, and because we originally intended to collect values for perceived uncertainty from English speakers (see Section 3.4.1), we translated all lexical uncertainty cues from German to English using DeepL as well as bilingual essays [81, 82].

3.2 Stage II: Collection of Non-Verbal Cues

In the previous stage, we not only selected verbal uncertainty cues that seemed suitable for a virtual agent, but also observed a large number of visual cues in the collected videos. These included eyebrow, mouth, head and eye movements as well as face touches – all features that have already been considered in classic research on human uncertainty behavior [71]. Despite these observations, the videos from Stage I were not directly usable for transferring facial expressions to a virtual agent. There are a number of reasons for this, many of which stem from the uncontrolled nature of the first study, including insufficient video quality, poor lighting, and occlusions (by a face mask, headset, hair, or face touches). We therefore followed up with a controlled lab study specifically designed to capture facial expressions as well as head and gaze data.

Table 1: Uncertainty cues selected after the data collections in Stage I and Stage II.

Filler Word	*none*, um, uh
Pre Hedge	*none*, maybe, I think
Post Hedge	*none*, but I'm not sure, I don't know
Intonation	*falling, rising*
Pre Pause	*Pause before the answer incl. pre hedge and filler*
Perform Length	*Length of the answer*
Post Pause	*Pause after the answer incl. post hedge*
Movement	*brow, lower face, eyes, head*

Pipeline Stage 2: Collecting Non-Verbal Cues

Input: None
Output: Four non-verbal uncertainty cues (see Table 1) captured on 191 facial video recordings
Task:
1 Record specific facial expressions of users for calibration.
2 Collect videos of users answering trivia questions.
3 Discard videos with non-answers.

3.2.1 Materials. In the second study, we used the same web app and trivia questions as in the first study. However, to remove the limitations of the first study that could be unfavorable for the animation of a virtual agent (e.g., lengthy answers or the visible embarrassment of participants), we revised the instructions (see Figure 2 in the supplementary materials). Since our focus is on the head of a virtual agent, we also explicitly excluded facial touches that would require animation of the agent's body (cf. requirement (3) in Section 3.1.4).

3.2.2 Methods. Participants were seated at a table with a laptop in front of a white wall and the experimenter ensured that nothing covered the participant's face. Participants were also instructed to keep their hands on the table throughout the study. Before beginning the quiz, participants were required to perform four calibration poses that focused on eyebrow movement and were intended to improve performance of the subsequent face capturing. After these calibration shots, the experimenter left the room and the procedure continued as described in Section 3.1.2.

3.2.3 Participants. 10 participants (5 women, 3 men, 2 non-binary) took part in the second phase of the data collection. The distribution by age group was as follows: 18-24 years ($N = 5$), 25-34 ($N = 4$), and 35-44 ($N = 1$).

3.2.4 Results. 294 response videos were recorded, of which 103 were marked as non-answers by the participants themselves. All participants adhered to the instructions so that no videos had to be removed due to occlusions.

3.3 Stage III: Cue Transfer to Virtual Agent

The next step of the pipeline was to transfer the identified verbal and non-verbal uncertainty behavior to a virtual agent. By synthesizing videos of a virtual agent instead of further using the original participant recordings of the two first stages, it was possible to exclude cues that cannot or are not intended to be simulated with the virtual agent in the final application (e.g., facial touches). We were also able to augment the training dataset by creating combinations of uncertainty cues that were not present in the initial participant recordings. Finally, the reliance on synthesized videos proved to be beneficial in terms of participant privacy.

Pipeline Stage 3: Transferring Uncertainty Cues to Agent

Input: Eight categories of (non-)verbal uncertainty cues; 191 facial video recordings of users
Output: 6476 synthesized agent videos with diverse uncertain behavior
Task:
1 Perform facial calibration for users.
2 Transfer facial movements in videos to virtual agent.
3 Synthesize speech from textual responses and uncertainty cues.
4 Render agent videos with combinations of facial animations and synthesized speech.

3.3.1 Agent Model. For the agent model, we used Reallusion's "Camila" [33], which is included in the Character Creator 4 software [32]. The model was imported into Unity 2021.3.6f1 using the Reallusion Auto Setup tools [31]. In order to maximize the transferability of the final open source ML model to the agent models of other researchers, we switched to the more limited CC4 Standard Expression Set, which consists of the standardized ARKit blendshapes. We further enabled the wrinkle system to increase the agent's expressiveness.

For animating Camila, we used the AccuFACE plugin for iClone 7 [30]. Supported by the NVIDIA RTX GPU, it enables capturing facial expressions from RGB videos.

3.3.2 Agent Facial Animation. In a preparatory step, a calibration was performed for each of the 10 participants of Stage II (see Section 3.2.2). Subsequently, all 191 response videos recorded in Stage II were manually transferred to Camila. For the lower face region (including mouth, cheeks and nose), we decided to remove movements that could be the result of vocalizing filler words or hedges. We therefore set the strength of 13 blendshapes to zero (*jawOpen/Forward/Left/Right, mouthFunnel, mouthPucker, mouthRollUpper/Lower, mouthPressLeft/Right, mouthStretchLeft/Right, tongueOut*). In contrast, the blendshape weights for movements such as mouth smile, dimple and frown were retained. Each of the 191 animations was manually divided into three sections - before, during and after the participant's response, which was determined via the audio channel. The resulting animations were then exported to Unity.

3.3.3 Agent Speech. For the virtual agent's responses, we discarded the quiz responses from the first two stages and replaced them with English first names. We selected the 15 most popular two-syllable first names for both male and female babies born in the United States between 1923 and 2022, according to the United States Social Security Administration [2]. The intention behind this choice was to use generic responses that could fit a variety of hypothetical questions and that do not inherently carry some uncertainty. For example, observers might ascribe a higher degree of certainty to the answer "Shakespeare" than to the answer "Kafka", even without knowing the question, simply because of the different familiarity of the two writers.

All 30 first names, as well as the filler words and pre/post hedges selected in Section 3.1.4, were synthesized using the Google Text-to-Speech service [24], with the *en-US-Neural2-F* model. Pre-hedges were recorded twice as the intonation of the following word affected their own pitch variation.

Based on the 191 animations × 3 filler word options (*none* / um / uh) × 3 pre hedge options (*none* / I think / maybe) × 3 post hedge options (*none* / I don't know / but I'm not sure) × 2 intonations, we initiated the rendering of 10,314 videos in Unity. Combinations were skipped if the audio files of the respective filler words and/or hedges were longer than the pre/post animation clips, resulting in a total of 6476 videos.

3.4 Stage IV: Annotation of Perceived Uncertainty

Since the ultimate goal of our project pipeline is to create agent behavior that conveys a specified level of uncertainty to the user, we were interested in the uncertainty perception of the observer rather than the speaker (perceptions that can differ according to Pon-Barry [61]). In Stage IV, we therefore let people rate the synthesized videos from the previous stage.

Pipeline Stage 4: Annotating Perceived Uncertainty

Input: Synthesized agent videos with diverse uncertain behavior
Output: 3000 subjective scores for the confidence expressed by the agent
Task:
1 Let users label each video with a perceived confidence score.
2 Assess quality of the video labels using a random forest regressor.
3 Estimate relevance of each uncertainty cue for the perception of confidence.

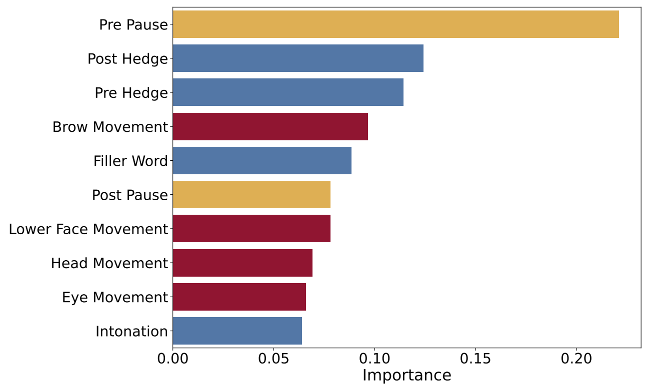

Figure 3: Feature importances of the verbal cues (blue), non-verbal cues (red), and pause lengths (yellow) as output by the random forest regressor. All values add up to 1.

3.4.1 Methods. The crowdsourcing platform Amazon Mechanical Turk was used for the initial labeling of all 6476 videos. Despite extensive precautions, subsequent analyses revealed that the collected data was mostly random (see the supplementary materials for a full description of this step including the analyses). For this reason, the data was discarded and a further study iteration with university members was conducted instead. Due to the smaller pool of participants compared to crowdsourcing, only 3000 videos were labeled. Each 120 videos were combined into one task set, whereby it was ensured that the length of the videos was roughly balanced between the task sets.

The data collection was designed as an online study, using LimeSurvey to obtain demographic data as well as a custom website and database hosted on a local server for the actual labeling process.

As in related studies, we omitted the original questions [9] and only showed the participants the response videos. Participants were instructed to enter a value between 0% (the virtual human appears to be completely uncertain) and 100% (the virtual human appears to be absolutely certain) into a numerical input field. They also had to select which answer the virtual human had given from a drop-down list with five possible answers. This step fulfilled several quality assurance purposes: (1) input was only possible after the participant had watched the entire video (ensured by a script), (2) correct input required that the participant had activated the audio, and (3) the task served as an attention check. Both the input field for entering a confidence value and the drop-down list have been embedded in a sentence (see the supplementary materials for the full instructions).

3.4.2 Participants. 24 participants (13 women, 11 men) took part in the labeling process, with one of them completing two task sets. Twelve participants were between 18 and 24 years old, nine between 25 and 34 years, two between 35 and 44 years, and one between 45 and 54 years. 22 participants stated their native language as German and two as Russian. Student participants ($N = 19$) received course credit for their participation.

3.4.3 Analysis and Results. Based on the Unity output and the labeling process, the resulting dataset contains combinations of different verbal and non-verbal uncertainty cues as well as a rating for the subjectively perceived confidence that they evoke in an observer when expressed by an agent.

A random forest regressor from the scikit-learn package [60] version 1.4.1 was fit to the data with default parameters. To verify whether participants adhered to a (possibly individual) standard when labeling the videos, we first normalized the data per participant. The regression model's predictions for a test subset of the data indicated a non-random labeling of videos (for a comparison with the MTurk labeling, see Figure 4 in the supplementary materials).

Another output of the regressor are the feature importances as depicted in Figure 3. For non-verbal cues, frame-to-frame differences of all involved blendshapes (for brows and the lower face region) or rotations (for head and eyes) were aggregated. The resulting values were divided by the number of frames to obtain features that are independent of the pre and post pause. As can be seen in the bar chart, the length of the pause before the agent's actual response was most indicative of the confidence estimation by the annotators. Overall, all of the considered uncertainty cues seem to have contributed to some extent to the perceived confidence.

In a follow-up analysis, it was investigated whether the combination of multimodal cues also improved the accuracy of the perceived uncertainty ratings (cf. [71]). In addition to the previously trained random regressor for all features, one random regressor each was fitted to the data for verbal, non-verbal and pause length features. The regressors were used to predict the confidence values for a test set as described above. Prior to comparing them with the corresponding observed confidence scores, the previously applied normalization per participant was reversed. This results in directly interpretable values in the original scale, with 0 and 100 indicating low and high confidence, respectively. Both non-verbal features (Pearson's $r = .621$, $RMSE = 21.713$) and pause lengths (Pearson's $r = .618$, $RMSE = 21.778$) yielded a slightly higher correlation and lower root mean square error on the test set than verbal features (Pearson's $r = .591$, $RMSE = 22.270$). Overall, there is a weaker correlation and a higher RMSE for all three feature groups when they are considered in isolation rather than in combination (Pearson's $r = .685$, $RMSE = 20.160$). The differences are depicted in Figure 5 in the supplementary materials.

3.5 Stage V: Training of an ML Model

In this stage, the previously compiled dataset is used to train an ML model that is intended to predict appropriate uncertainty cues from a given confidence score. To generate both animation (i.e., blendshape and head/eye rotation) sequences and verbal cues, such as fillers, hedges, and intonation, we utilize a combination of two interconnected deep-learning modules as shown in Figure 4.

3.5.1 Metadata. For our ML model to learn the metadata, including the verbal cues and sequence lengths (cf. Table 1), we fine-tuned a 7-billion parameter LLaMA-2 instruction model [74] using low rank adaptation (LoRA) [28]. LLaMA-2 is a transformer-based [76] LLM for text generation that was trained using a GPT approach [63].

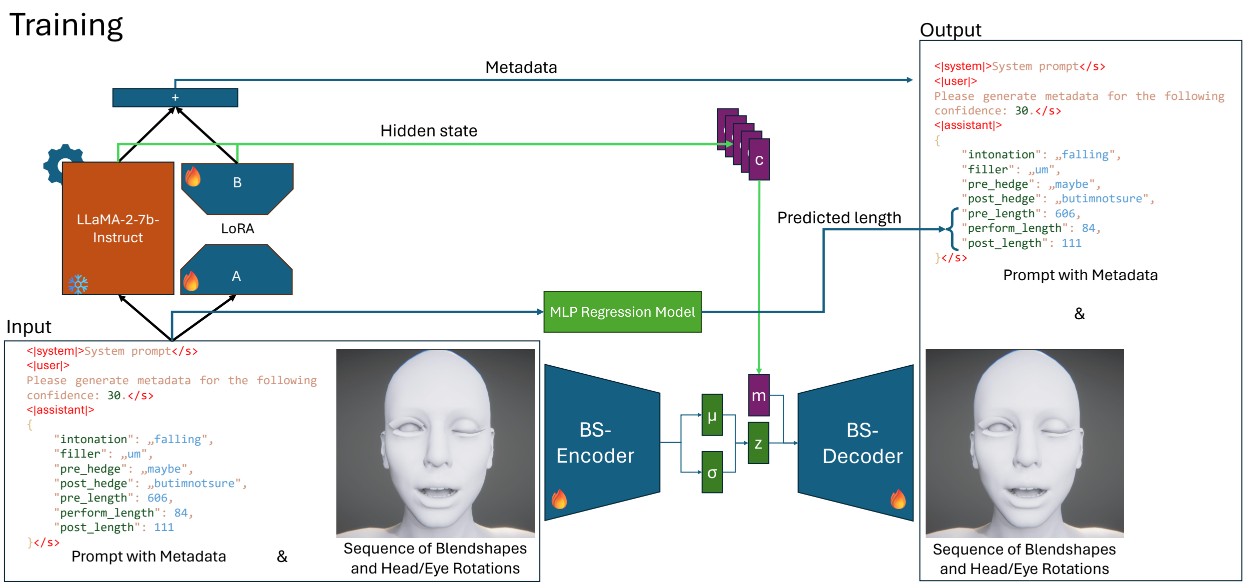

Figure 4: Pipeline used to train our deep-learning architecture, described in Section 3.5. For the subsequent generation of uncertainty behavior, we utilize the inference capabilities of a 7b LLaMA-2 instruction model [74] with the decoder part of our model (see Figure 6 in supplementary materials). The fire symbol indicates trained weights, while the snowflake symbolizes frozen weights. Agent model by Reallusion Inc.

Pipeline Stage 5: Model Training

Input: Training dataset comprised of

- 191 × 3 text files with pre-/mid-/post-response animation sequences for 54 blendshape weights as well as the orientation of the head and eyes
- 3000 JSON files with the properties *intonation*, *fillerWord*, *preHedge*, *postHedge*, *prePause*, *performLength*, *postPause* (= the "metadata"), and a reference to the *animation sequence*
- 1 confidence score for each JSON file

Output: ML model that predicts both verbal and non-verbal uncertainty cues from a single confidence value

Task:

1 Generate prompts from metadata and confidence values.
2 Train LLaMA-2 instruct for verbal cues, a multi-layer perceptron for pre/post pauses and perform length, and a variational autoencoder for non-verbal cues.
3 Generate novel metadata and animation sequences through the trained models.

As the instruction model expects a system and user prompt to generate a response, we decided on a system prompt that explains the minimum and maximum confidence values along with a brief description of the task (cf. supplementary materials Figure 7). As the user prompt, we provide a short request to generate metadata for a provided confidence value and expect an output string in JavaScript Object Notation (JSON) with all uncertainty cues listed in Table 1 except for movements. As the pre pause was found to be a highly determining factor for the perceived uncertainty (cf. Figure 3) and intermediate tests showed that it is only insufficiently modeled through the generative prediction process of LLaMA-2, we decided against using this approach for all numerical values of our metadata. Reasons for the poor accuracy could be both our own small training dataset size and previous findings that the performance of pre-trained language models on numerical reasoning tasks drops when the numerical terms of interest are less frequent in the pre-trained data [64]. The latter is likely the case with our sequence lengths, which are often greater than 100. We therefore train a small multi-layer perceptron (MLP) model to predict the pre and post pause from the other uncertainty cues as well as the confidence value, and combine the processes of the instruction and the MLP model to generate the full metadata.

For training and inference, we generate prompts for all samples collected in Stages I to IV and fine-tune our LLaMA-2 model using LoRA based on these. Figure 4 shows our prompts that keep the textual character of numerical values, which has been shown to work successfully in other related tasks [26]. When generating novel samples, we can then prompt the model by simply providing the start token with a confidence value to produce the remaining metadata.

3.5.2 Blendshapes. As our second module, we used a Conditional Variational Autoencoder (CVAE) [69] architecture for sampling novel blendshape sequences. CVAEs are Variational Autoencoders (VAE) [39] that allow the generation of novel samples based on a variational Bayesian inference model [47] by learning the prior and posterior of the data distribution. Most commonly, a CVAE is constructed of an encoder, decoder, and latent space representation [39, 47]. In our case, we utilized convolutional encoders and decoders to process the blendshape sequences. To provide the condition, we averaged the output of the last hidden layer of LLaMA-2 and concatenated the generated context vector with the latent space to feed them into the decoder. During inference, we also smoothed the generated blendshape sequences for the head using a Savitzky–Golay filter with the length of 101 with polynomial order of 3 and utilized a length of 31 for the eyebrows.

3.5.3 Training & Hyperparameters. To train the model, we calculated the mean square error (MSE) between input and generated output. As suggested by Kingma and Welling [39], we employed the re-parameterization trick and computed the Kullback-Leibler divergence for our latent loss. For the metadata, we calculated the cross entropy loss as used by Radford et al. [63]. We trained LLaMA-2 for 5 epochs and the CVAE for 1000 epochs. The training of our LLaMA-2 model was performed through LoRA, with parameters of $r = 32$, $\alpha = 64$, and a dropout of 0.1. We only stored the model with the best reconstruction loss on our validation dataset. During training, we set the learning rate to 0.0003, the batch size to 2 accumulating over 10 batches. Further, we used AdamW [51] to perform the optimization process using a weight decay of 0.001 for all parameters except the bias and layer normalization weights. For the MLP regression model, we used 3 layers with a hidden layer size of 16 and a tanh activation function. We trained the regression model through a mean square error loss for 60 epochs using a learning rate of 0.001 and a L2 regularization term of 0.0001.

3.5.4 Generation of Novel Samples. To generate novel samples (cf. Figure 6 in the supplementary materials for the inference architecture), we utilize the generative abilities of LLaMA-2. Unlike in training, we do not provide the LLM with the metadata, but only with the system and user prompt, and generate the rest of the response. For the user prompt, we replace the confidence value with the confidence we want to generate our metadata for. For the generation, we employ probabilistic multinomial sampling with a temperature of 1.1 and a token probability of 98%. Afterward, we estimate the pre pause and post pause from the generated metadata and feed the adapted metadata description back into LLaMA-2 to extract the context from the last hidden state. We pass this last hidden state to the CVAE as the conditional variable, the same as during our training. However, we omit the encoder part of our CVAE and draw from a normal distribution to sample from the embedding space, generating novel sequences that are dependent on the generated metadata.

3.5.5 Analysis and Results. To get an overview of the generated metadata and sequences as a function of the input uncertainty, we considered 100 outputs of the ML model at each of 11 uncertainty levels (0% to 100%, in 10% steps). For all individual uncertainty cues, relative frequencies (e.g., for filler words and intonation) or mean values (e.g., for pre pause and brow movement) can be found in the supplementary materials (Figures 8–13).

To our knowledge, there are no models that generate both verbal and non-verbal cues from an uncertainty value to compare our model with. To demonstrate why we chose a pipeline of dedicated models for predicting pause lengths (regressor), verbal cues (LLaMA-2), and non-verbal cues (CVAE) rather than using an encompassing LLM, we evaluated against LLaMA-2. We converted the blendshape sequences into byte sequences using singular bytes as tokens and trained the model using cross entropy. Current LLMs, like LLaMA-2 and GPT-4o, are limited by output length (4096 tokens) and memory requirements, making them impractical for generating long sequences of blendshapes. We require approximately 200 tokens for metadata, including verbal cues, and 212 tokens (41 blendshapes with 4 bytes each plus one head orientation and two eye orientations with 16 bytes each) per frame for non-verbal cues. Therefore, LLaMA-2 only allows $(4096 - 200)/212 = 18$ frames in the context. Hardware limitations forced us to reduce the length to 1472 tokens, accommodating 6 frames. With an overlap of one frame, predicting 30 seconds of agent behavior with 60 frames per second requires 360 generation calls to LLaMA-2. Therefore, the process of just predicting a single sequence is very time-consuming (3.30 hours[2]), with an energy consumption of $0.85kWh$ producing $420g$ CO_2. In comparison, our model takes 4.0 seconds[2] per sequence ($\approx 0.003kWh$, $\approx 0.15g$ CO_2). Similarly, our model requires only about 0.1% of the computational power of LLaMA-2, both for training and for generating sequences (see Table 5 in the supplementary materials for the exact values).

With regard to the quality of the resulting uncertainty cues, we compared the pause lengths and non-verbal cues generated with our pipeline (i.e., with a regressor and a CVAE) to those generated with LLaMA-2. For the LLaMA-2 baseline model, we discarded the MLP regression layer of our final model.

We generated 1100 samples each with our regressor and the LLaMA-2 baseline model (100 for each confidence level between 0% and 100% in steps of 10%). For the baseline model, one of the generated samples was an extreme outlier with a pre pause of 1092.25 seconds. After removing this outlier and averaging the generated pause lengths over each certainty level, the error distance between training and generated data was similar for the regressor ($RMSE = .940$) and the LLaMA-2 baseline model ($RMSE = .955$). However, considering the distribution of the generated pause lengths (see supplementary materials Figure 12), it becomes apparent that the LLaMA-2 baseline model only predicts values around the mean for all certainty levels. In comparison, the regressor picks up the trend of decreasing pause lengths with increasing certainty that was present in the training data.

As can be seen in the supplementary video, the 360 calls to LLaMA-2 that are required to predict 30 seconds of agent behavior generate disconnected blendshape snippets, as the third call has no knowledge of the blendshapes generated in the first call. We believe these visual examples better showcase the blendshape quality compared to traditional metrics like RMSE. Our CVAE was designed to generate unique sequences for the same uncertainty input, which might result in lower metric performance but subjectively more natural animations.

3.6 Stage VI: User Evaluation

Since the common ML error metrics have only limited expressiveness regarding the naturalness of the ML output, we evaluated the fidelity of the generated agent responses in a user study.

[2]On a server with 8x Nvidia RTX A6000, 2x AMD EPYC 7302, and 256 GB RAM.

(a) 80% confidence; Uh... I think ___?; pause = 2.3sec **(b) 10% confidence; Um... I think ___? I don't know.; pause = 5.5sec** **(c) 20% confidence; Um... ___. I don't know.; pause = 3.5sec** **(d) 10% confidence; Um... I think ___? I don't know.; pause = 5.5sec**

Figure 5: Examples of subtle facial indicators of uncertainty that our trained ML model was able to reproduce: (a) raised eyebrows and diverted gaze, (b) additionally dropped corners of the mouth, (c) contracted eyebrows, and (d) raised inner eyebrows. Some facial expressions were concatenated within one sequence, such as (b) and (d). Agent models by Reallusion Inc.

Pipeline Stage 6: Evaluation of the ML Model

Input: 44 configurations of uncertainty cues generated by the ML model

Output: Statistics on subjective scores for the confidence expressed by the agent

Task:
- Apply configurations to four different agent models.
- Let users label each configuration with a perceived confidence score.
- Calculate differences between perceived and predicted confidence values.
- Analyze effects of factors such as the agent model, as well as intra- and inter-rater reliability.

3.6.1 Materials. Using the inference pipeline described in Section 3.5.4, we created 44 responses, four for each confidence value between 0% and 100% in steps of 10%. Each of the 44 configurations was applied to four agent models representing different genders, age groups and ethnicities (see Figure 5). As for the training of the ML model, we used the Reallusion character Camila. The other three models are also provided by Reallusion and were dubbed using Google text-to-speech, more specifically Alisha (en-US-Neural2-G), Kevin (en-US-Neural2-D), and Susan (en-US-Neural2-E) [33, 34]. All videos were rendered in Unity.

3.6.2 Methods. The protocol for the user evaluation was the same as for the annotation of agent videos when collecting training data for the ML model (see Section 3.4.1). The only difference in the procedure was the number and type of agent videos for which the participants had to rate the perceived confidence. Each participant rated all 44 configurations, whereby the assignment to the agent models differed between participants, but was balanced. In total, each participant was assigned one configuration for each confidence level and each agent. After all 44 videos had been shown, 11 of them were presented to the participants a second time. These videos were used to investigate possible variability in a participant's ratings. The 11 videos covered the entire confidence range and were identical to videos 12 to 22, but in a shuffled order. By choosing the second block of videos, we aimed to both reduce the impact of a possible burn-in phase, in which participants might still develop internal guidelines for rating the videos, and to create a gap between the first and second playback so that participants would not notice the repetition and be unable to recall their previously given ratings. Study participants were instructed with the same task as in Section 3.4.1.

3.6.3 Participants. 12 participants took part in the online evaluation (4 women, 7 men, 1 non-binary). Their reported age ranges were 18-24 ($N = 3$), 25-34 ($N = 8$), and 35-44 ($N = 1$). 8 of our participants stated that German was their native language, 3 Chinese and 1 French. We ensured that none of them participated in the annotation of training videos in Stage IV to avoid bias.

3.6.4 Analysis and Results. Figure 6 shows the results of the evaluation along with the standard deviation and regression line. Our model shows a trend of higher confidences being perceived as such. However, we could not reproduce the full range of uncertainty through our model. A discussion of possible reasons follows in the next section. Overall, the mean absolute difference between the input confidence and the rated perceived confidence was 25.005 ($SD = 19.074$).

Variability in Confidence Ratings We observed a high variability in the confidence scores for each input confidence level. This could be an indication of some imprecision in the trained ML model, i.e., uncertainty cues might be predicted that do not precisely represent the input confidence level. However, the result could also be due to other factors, such as the same predicted agent video being perceived differently by multiple participants or even by the same participant on repeated evaluation. In this case, the reason for the high variability would be the complexity of the task of judging the

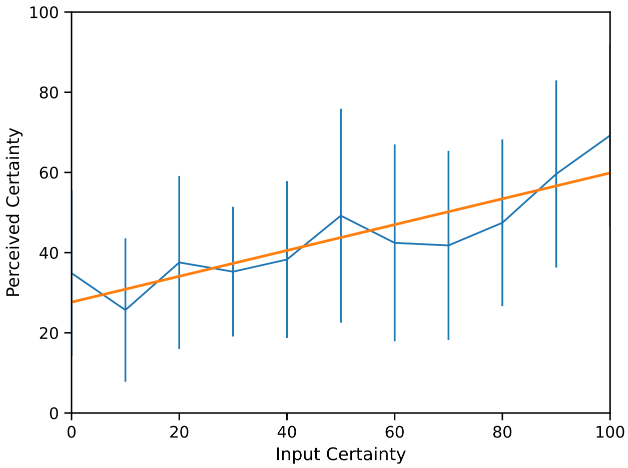

Figure 6: *(blue)* Line plot showing the certainty estimated in the user evaluation (cf. Section 3.6) in comparison to ground truth. The vertical error bars depict the standard deviation. *(orange)* A linear regression model was fit to the data.

confidence of an agent video rather than the ML model. To further investigate this relationship, we calculated the reliability, i.e. the degree of correlation and agreement, both between and within the participants' confidence ratings using the intra-rater correlation coefficient method of SPSS, version 27.

We first calculated each participant's intra-rater reliability, which expresses how self-consistent the participant is in their scoring of videos. ICC estimates were calculated based on a single-rater, absolute-agreement, two-way mixed effects model. The resulting intra-rater reliability varied greatly from individual to individual, ranging from .135 with 95% CI (-.556, .673) to .921 with 95% CI (.733, .978). The values for all participants can be found in the supplementary materials (Table 6). For inter-rater reliability, which reflects the variability in the perception of uncertainty between participants, we used a single-rater, absolute-agreement, one-way random effects model. The ICC(1,1) was estimated to be .421 with 95% CI (.339, .501). These values indicate a poor to moderate inter-rater reliability [42].

Generalizability to Different Agent Models Despite differences in appearance, facial morphology and voice, all four tested agent models were rated similarly in their expressed confidence. The mean values of the difference *perceived confidence – input confidence* ranged from -5.423 (SD = 14.191) for Camila to -7.220 (SD = 11.639) for Kevin.

For analyzing the data, we aggregated all confidence differences per participant and agent. As their residuals were not normally distributed according to a Shapiro-Wilk test and visual inspection of the Q-Q-plot, we performed a Friedman test. No statistically significant effect of agent model on difference between perceived confidence and input confidence was found ($\chi^2(3) = 1.300, p =$ .729).

While the analysis is based on only 12 participants, an examination of the plotted data across all confidence levels (see Figure 14 in the supplementary materials) supports the assumption that there was no systematic over- or underestimation of the confidence of one of the agents in the tested setup. This would be in line with the results of Brennan and Williams [9], who found that the feeling of another's knowing was not affected by social stereotypes evoked by different voices or gender.

4 LIMITATIONS AND DISCUSSION

Fidelity While we found that perceived certainty was increasing with rendered certainty, we observed a large standard deviation for certainty scores in the final user evaluation. One possible contributing factor to our results is the complexity of the problem at hand. While familiarizing ourselves with the data, we found that even short subtle facial expressions can change the overall certainty perception, even though all other features (words, intonation, pause length) remain constant. For example, a short, otherwise certain answer with no filler words or hedges can appear to be a random guess if it is followed by an uncertain pose with raised eyebrows and dropped corners of the mouth. Learning all these subtle interplays of features could require a much larger training dataset, preferably with multiple annotations per video.

In comparison to Oh and Stone [57], who found a mean absolute error (MAE) of 15.310 between self-reported and perceived uncertainty ratings, the overall error for our ML model was a little higher (MAE = 25.005). Since their agent was highly simplified with manually created animations, it may have omitted complex interactions of multiple features. On the other hand, as pointed out by the same authors [70], this manual approach has its own limitations. If a more realistic range of facial expressions should be reproduced, the higher number of blendshapes can no longer be combined independently. Our ML approach automatically learns which blendshapes can be animated simultaneously without causing distorted facial expressions.

By automating the generation of uncertain non-verbal behavior, our resulting animations exhibit a certain amount of undesirable micro-movements, as can be seen in the supplementary video. Nonetheless, the results of our proof of concept are very promising, considering that we only had a limited amount of training data available to produce the blendshapes. For a practically used model, larger amounts of training data would have to be collected or the CVAE could be pre-trained using other data sets that do not include uncertain behavior.

Another observation with respect to the data presented in Figure 6 is the overestimation and underestimation at low and high confidence values, respectively. This leads to a reduction in the total range of confidence that can be represented. This compression occurs naturally due to floor/ceiling effects and the averaging of confidence ratings, with an amplification due to the observed low intra- and inter-rater reliability. While the effect was also present in previous results with manually animated uncertainty behavior [52, 57], it might have been further increased by the ML model. Our analyses show that the model has correctly learned that, for example, the frequency of hedges should decrease with increasing confidence. Nevertheless, at 100% confidence, the post hedge “I don't know” is occasionally predicted (in 4 out of 100 cases), although it is usually not associated with a person being perfectly certain. A larger amount of training data could also lead to an improvement here.

Multimodality Our analysis presented in Section 3.4.3 indicates that for embodied agents (1) both verbal and non-verbal cues contribute to uncertainty perception, and (2) their combination leads to a more consistent perception. These results suggest that previous findings from human-human interaction, which demonstrated an improvement in the accuracy of perceived uncertainty ratings when combining multiple cue modalities [71], can be transferred to human-agent interaction. It should be noted that our study design only allows conclusions for human-agent interactions with an embodied agent, as verbal features were never presented in isolation.

We further found that some cues (e.g., brow movements and hedges) are better predictors of uncertainty than others (e.g., head movements and filler words). The length of the pause before the agent's actual response had the highest feature relevance. While such a pause could also be implemented in a chatbot or a voice-based agent, where it would represent a simple delay, our study only involved embodied agents that usually filled pauses with facial movements. Therefore, it can be hypothesized that the high relevance of pause length is greatly related to the non-verbal feedback and not only the delay. Future studies directly comparing embodied agents and purely text- or voice-based agents could determine the exact contributions.

Selection of Uncertainty Cues In our collection of (non-)verbal uncertainty cues, we observed a complex, partially subtle uncertainty behavior with a richness of (multimodal) cues that are beyond the scope of this project. We therefore selected a subset of uncertainty features based on the criteria presented in Section 3.1.4. Coders with different backgrounds might make different choices during the coding process, which is why, following the best practice of qualitative research, we specify that the coder has a background in human-computer interaction [56]. Further extensive studies are warranted to increase the richness of uncertainty behavior of virtual agents. For example, current LLMs could offer more diverse and natural verbal expressions for uncertain behavior when prompted accordingly.

Influence of Language and Culture All data collection steps were conducted with native German speakers and lexical uncertainty cues were translated into English based on bilingual studies (to support transparency and replicability, we have provided both German and English codes in the supplementary materials). Subtle differences in meaning may affect the frequency of observed uncertainty cues. In addition, Doupnik and Richter [19] have shown that the interpretation of uncertainty terms differs between cultures (in this case Germany and the US) not only due to translation, but also due to a language-culture effect. To counteract this, we had planned to limit the annotation of our videos to MTurk workers from the US. However, due to the low quality of the collected data, we had to have the videos re-annotated by members of the local university. Therefore, the ML model was trained on the perception of uncertain (English) behavior primarily by German speakers. Cultural differences, not only in the interpretation but also in the expression of uncertainty, need to be further explored to ensure high usability of uncertain virtual agents for diverse users.

Context Dependency Following related projects, we embedded our data collection in a quiz. This setting could elicit a different response behavior than a real question, for example asked by a colleague, would. In Stage I of the data collection, we observed several instances of embarrassed behavior (e.g., giggling), but this seemed to be mitigated by revising the instructions in Stage II. Nevertheless, reactions in a particular setting might differ from everyday conversations, where it is usually assumed that the questioner does not know the answer to a question themselves and an answer might have a higher significance. Furthermore, there might be contexts or tasks where a precise uncertainty statement is preferable to a natural answer, for example when asking a virtual agent about the probability of rain. It would be interesting to compare user perceptions for precise and natural virtual agents in different contexts in subsequent empirical studies.

Individual Differences We found individual differences both in the expression of uncertainty (e.g., preference for specific terms, length of response, expressiveness of the face) and in the interpretation of uncertain behavior (e.g., tendency of annotators to score all videos rather high or low, reflected in the poor to moderate inter-rater correlation). These justify future ML models that calibrate to interaction partners, the discourse and the shared context [43]. Instead of attempting to create a universal model, it might be interesting to consider both the personality of the user and a depicted personality of the agent.

5 CONCLUSIONS

In this paper, we presented our novel approach to convey to the user the otherwise hidden features of an ML model by leveraging the potential of embodied intelligent virtual agents. We explored the idea of virtual agents expressing the uncertainty of the underlying ML model through verbal (e.g., filler words, intonation) and non-verbal (e.g., raised eyebrows, upward gaze) behavior. To investigate the feasibility of our approach, we employed a full pipeline from observing human uncertainty behavior, to training our ML model to emulate this behavior in virtual agents, to a final human-based evaluation. While we found positive evidence that current virtual agents are capable of exhibiting complex uncertainty behavior, we also found that it is demanding for (1) an ML model to learn all subtle interplays of cues and (2) for users to distinguish narrow ranges of uncertainty. We have discussed various approaches to address these potentials in future developments, both in terms of the model architecture and human factors such as the personality, culture, and context of the user.

In conclusion, while lacking the precision required for experts, embodied agents may be an intuitive alternative to traditional visualizations for non-experts. With their increasing ubiquity (cf. GPT-4o, Metaverse), our research underpins the importance of a close coupling of verbal and non-verbal signals. Not only was nonverbal feedback an effective predictor of uncertainty, but consistent judgment of an agent was more difficult when relying solely on either modality. We observed how a brief frown can turn a seemingly certain message into a random guess. This requires a carefully coordinated timing of cues that goes beyond the manual definition of key behaviors described in previous literature [70]. Our ML architecture could also be applied to similarly complex communication aspects that involve non-verbal feedback, like emotions, attitude, personality, or attention. Overall, this proof of concept is intended to contribute to the research into the transparency of ML models. At the same time, it could serve as an example of the combination of ML and virtual reality that could be applied to other areas such as explainability and data privacy.

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REFERENCES

[1] Moloud Abdar, Farhad Pourpanah, Sadiq Hussain, Dana Rezazadegan, Li Liu, Mohammad Ghavamzadeh, Paul Fieguth, Xiaochun Cao, Abbas Khosravi, U Rajendra Acharya, et al. 2021. A review of uncertainty quantification in deep learning: Techniques, applications and challenges. *Information fusion* 76 (2021), 243–297.

[2] Social Security Administration. 2024. Top Names Over the Last 100 Years. https://www.ssa.gov/oact/babynames/decades/century.html (Accessed January 6, 2024).

[3] Sebastian Andersson, Kallirroi Georgila, David Traum, Matthew Aylett, and Robert AJ Clark. 2010. Prediction and realisation of conversational characteristics by utilising spontaneous speech for unit selection. In *Proceedings of Speech Prosody*.

[4] Marjolijn L Antheunis, Alexander P Schouten, Patti M Valkenburg, and Jochen Peter. 2012. Interactive uncertainty reduction strategies and verbal affection in computer-mediated communication. *Communication Research* 39, 6 (2012), 757–780.

[5] Emma Beauxis-Aussalet, Elvira Arslanova, Lynda Hardman, and Jacco Van Ossenbruggen. 2013. A case study of trust issues in scientific video collections. In *Proceedings of the ACM international workshop on multimedia analysis for ecological data*. 41–46.

[6] Emma Beauxis-Aussalet, Joost van Doorn, and Lynda Hardman. 2019. Supporting End-User Understanding of Classification Errors: Visualization and Usability Issues. *The Journal of Interaction Science* 7 (2019), 1–29.

[7] Simon Betz, Birte Carlmeyer, Petra Wagner, and Britta Wrede. 2018. Interactive hesitation synthesis: modelling and evaluation. *Multimodal Technologies and Interaction* 2, 1 (2018), 9.

[8] Umang Bhatt, Javier Antorán, Yunfeng Zhang, Q Vera Liao, Prasanna Sattigeri, Riccardo Fogliato, Gabrielle Melançon, Ranganath Krishnan, Jason Stanley, Omesh Tickoo, et al. 2021. Uncertainty as a form of transparency: Measuring, communicating, and using uncertainty. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. 401–413.

[9] Susan E Brennan and Maurice Williams. 1995. The feeling of another's knowing: Prosody and filled pauses as cues to listeners about the metacognitive states of speakers. *Journal of memory and language* 34, 3 (1995), 383–398.

[10] John S Bridle. 1990. Probabilistic interpretation of feedforward classification network outputs, with relationships to statistical pattern recognition. In *Neurocomputing: Algorithms, architectures and applications*. Springer, 227–236.

[11] Wibecke Brun and Karl Halvor Teigen. 1988. Verbal probabilities: Ambiguous, context-dependent, or both? *Organizational Behavior and Human Decision Processes* 41, 3 (1988), 390–404.

[12] Kirill Bykov, Marina M-C Höhne, Klaus-Robert Müller, Shinichi Nakajima, and Marius Kloft. 2020. How much can I trust you? – Quantifying uncertainties in explaining neural networks. *arXiv preprint arXiv:2006.09000* (2020).

[13] Michelle Chua, Doyun Kim, Jongmun Choi, Nahyoung G Lee, Vikram Deshpande, Joseph Schwab, Michael H Lev, Ramon G Gonzalez, Michael S Gee, and Synho Do. 2023. Tackling prediction uncertainty in machine learning for healthcare. *Nature Biomedical Engineering* 7, 6 (2023), 711–718.

[14] Sunhao Dai, Yuqi Zhou, Liang Pang, Weihao Liu, Xiaolin Hu, Yong Liu, Xiao Zhang, and Jun Xu. 2023. Llms may dominate information access: Neural retrievers are biased towards llm-generated texts. *arXiv preprint arXiv:2310.20501* (2023).

[15] Fiorella De Rosis, Catherine Pelachaud, Isabella Poggi, Valeria Carofiglio, and Berardina De Carolis. 2003. From Greta's mind to her face: modelling the dynamics of affective states in a conversational embodied agent. *International journal of human-computer studies* 59, 1-2 (2003), 81–118.

[16] DeepL. 2024. DeepL Translate: The world's most accurate translator. https://www.deepl.com/en/translator/ (Accessed May 6, 2022).

[17] Cüneyt Demir. 2018. Hedging and academic writing: an analysis of lexical hedges. *Journal of language and linguistic studies* 14, 4 (2018), 74–92.

[18] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).

[19] Timothy S Doupnik and Martin Richter. 2003. Interpretation of uncertainty expressions: a cross-national study. *Accounting, Organizations and Society* 28, 1 (2003), 15–35.

[20] Mirta Galesic and Rocio Garcia-Retamero. 2010. Statistical numeracy for health: a cross-cultural comparison with probabilistic national samples. *Archives of internal medicine* 170, 5 (2010), 462–468.

[21] Jakob Gawlikowski, Cedrique Rovile Njieutcheu Tassi, Mohsin Ali, Jongseok Lee, Matthias Humt, Jianxiang Feng, Anna Kruspe, Rudolph Triebel, Peter Jung, Ribana Roscher, et al. 2023. A survey of uncertainty in deep neural networks. *Artificial Intelligence Review* 56, Suppl 1 (2023), 1513–1589.

[22] Tirthankar Ghosal, Kamal Kaushik Varanasi, and Valia Kordoni. 2022. HedgePeer: a dataset for uncertainty detection in peer reviews. In *Proceedings of the ACM/IEEE Joint Conference on Digital Libraries*. 1–5.

[23] Daena J Goldsmith. 2001. A normative approach to the study of uncertainty and communication. *Journal of communication* 51, 3 (2001), 514–533.

[24] Google. 2024. Text-to-Speech AI: Lifelike Speech Synthesis | Google Cloud. https://cloud.google.com/text-to-speech (Accessed January 6, 2024).

[25] Miriam Greis, Thorsten Ohler, Niels Henze, and Albrecht Schmidt. 2015. Investigating representation alternatives for communicating uncertainty to non-experts. In *Proceedings of the IFIP TC13 International Conference on Human-Computer Interaction (INTERACT)*. Springer, 256–263.

[26] Nate Gruver, Marc Finzi, Shikai Qiu, and Andrew Gordon Wilson. 2023. Large language models are zero-shot time series forecasters. *arXiv preprint arXiv:2310.07820* (2023).

[27] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778.

[28] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685* (2021).

[29] Jessica Hullman, Xiaoli Qiao, Michael Correll, Alex Kale, and Matthew Kay. 2018. In pursuit of error: A survey of uncertainty visualization evaluation. *IEEE transactions on visualization and computer graphics* 25, 1 (2018), 903–913.

[30] Reallusion Inc. 2024. AccuFACE - Video-based AI Facial Mocap | iClone. https://mocap.reallusion.com/iclone-motion-live-mocap/accuface.html (Accessed November 23, 2023).

[31] Reallusion Inc. 2024. Auto Setup for Unity | Character Creator. https://www.reallusion.com/auto-setup/unity/default.html (Accessed August 31, 2023).

[32] Reallusion Inc. 2024. Character Creator: 3D Character Design Software. https://www.reallusion.com/character-creator/ (Accessed August 31, 2023).

[33] Reallusion Inc. 2024. Free Assets for Animated Characters | Character Creator. https://www.reallusion.com/character-creator/3d-content.html (Accessed July 22, 2024).

[34] Reallusion Inc. 2024. Scan Reality - Alisha - Character Creator/Actor - Reallusion Content Store. https://www.reallusion.com/contentstore/pack/scan-reality-alisha (Accessed November 23, 2023).

[35] Susan Joslyn and Sonia Savelli. 2021. Visualizing uncertainty for non-expert end users: The challenge of the deterministic construal error. *Frontiers in Computer Science* 2, 590232 (2021).

[36] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221* (2022).

[37] Thashmee Karunaratne and Adenike Adesina. 2023. Is it the new Google: Impact of ChatGPT on students' information search habits. In *Proceedings of the ACI European conference on e-Learning (ECEL)*.

[38] Kangsoo Kim, Celso M de Melo, Nahal Norouzi, Gerd Bruder, and Gregory F Welch. 2020. Reducing task load with an embodied intelligent virtual assistant for improved performance in collaborative decision making. In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 529–538.

[39] Diederik P Kingma and Max Welling. 2013. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114* (2013).

[40] Ambika Kirkland, Harm Lameris, Eva Székely, and Joakim Gustafson. 2022. Where's the uh, hesitation? The interplay between filled pause location, speech rate and fundamental frequency in perception of confidence. In *Proceedings of the INTERSPEECH conference*. 4990–4994.

[41] Benjamin Kompa, Jasper Snoek, and Andrew L Beam. 2021. Second opinion needed: communicating uncertainty in medical machine learning. *NPJ Digital Medicine* 4, 1 (2021), 4.

[42] Terry K Koo and Mae Y Li. 2016. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of chiropractic medicine* 15, 2 (2016), 155–163.

[43] Stefan Kopp and Nicole Krämer. 2021. Revisiting human-agent communication: The importance of joint co-construction and understanding mental states. *Frontiers in Psychology* 12, 580955 (2021).

[44] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. *arXiv preprint arXiv:2302.09664* (2023).

[45] Nhat Le, Khanh Nguyen, Quang Tran, Erman Tjiputra, Bac Le, and Anh Nguyen. 2023. Uncertainty-aware label distribution learning for facial expression recognition. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 6088–6097.

[46] Yaniv Leviathan and Yossi Matias. 2018. Google Duplex: An AI system for accomplishing real-world tasks over the phone. (2018).

[47] Pengzhi Li, Yan Pei, and Jianqiang Li. 2023. A comprehensive survey on design and application of autoencoder in deep learning. *Applied Soft Computing* (2023), 110176.

[48] Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. Teaching models to express their uncertainty in words. *arXiv preprint arXiv:2205.14334* (2022).

[49] Zhen Lin, Shubhendu Trivedi, and Jimeng Sun. 2023. Generating with confidence: Uncertainty quantification for black-box large language models. *arXiv preprint arXiv:2305.19187* (2023).

[50] Isaac M Lipkus, Greg Samsa, and Barbara K Rimer. 2001. General performance on a numeracy scale among highly educated samples. *Medical decision making* 21, 1 (2001), 37–44.

[51] Ilya Loshchilov and Frank Hutter. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).

[52] Erwin Marsi and Ferdi Van Rooden. 2007. Expressing uncertainty with a talking head in a multimodal question-answering system. In *Proceedings of the Workshop on Multimodal Output Generation (MOG)*. 105–116.

[53] Sabrina J Mielke, Arthur Szlam, Emily Dinan, and Y-Lan Boureau. 2022. Reducing conversational agents' overconfidence through linguistic calibration. *Transactions of the Association for Computational Linguistics* 10 (2022), 857–872.

[54] Scott Monteith, Tasha Glenn, John R Geddes, Peter C Whybrow, Eric Achtyes, and Michael Bauer. 2023. Artificial intelligence and increasing misinformation. *The British Journal of Psychiatry* (2023), 1–3.

[55] Clifford Nass and Youngme Moon. 2000. Machines and mindlessness: Social responses to computers. *Journal of social issues* 56, 1 (2000), 81–103.

[56] Benjamin John Newton, Zuzana Rothlingova, Robin Gutteridge, Karen LeMarchand, and Jon Howard Raphael. 2012. No room for reflexivity? Critical reflections following a systematic review of qualitative research. *Journal of Health Psychology* 17, 6 (2012), 866–885.

[57] Insuk Oh and Matthew Stone. 2007. Understanding RUTH: Creating believable behaviors for a virtual human under uncertainty. In *Proceedings of the International Conference on Digital Human Modeling (ICDHM)*. Springer, 443–452.

[58] Walter J Ong and John Hartley. 2013. *Orality and literacy.* Routledge.

[59] Lace Padilla, Matthew Kay, and Jessica Hullman. 2020. Uncertainty visualization. (2020).

[60] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. 2011. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12 (2011), 2825–2830.

[61] Heather Pon-Barry. 2008. Prosodic manifestations of confidence and uncertainty in spoken language. In *Proceedings of the Annual Conference of the International Speech Communication Association.*

[62] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*. PMLR, 28492–28518.

[63] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. 2019. Language models are unsupervised multitask learners. *OpenAI blog* 1, 8 (2019), 9.

[64] Yasaman Razeghi, Robert L Logan IV, Matt Gardner, and Sameer Singh. 2022. Impact of pretraining term frequencies on few-shot reasoning. *arXiv preprint arXiv:2202.07206* (2022).

[65] Ian T Ruginski, Alexander P Boone, Lace M Padilla, Le Liu, Nahal Heydari, Heidi S Kramer, Mary Hegarty, William B Thompson, Donald H House, and Sarah H Creem-Regehr. 2016. Non-expert interpretations of hurricane forecast uncertainty visualizations. *Spatial Cognition & Computation* 16, 2 (2016), 154–172.

[66] Susanne Schmidt, Gerd Bruder, and Frank Steinicke. 2018. Effects of embodiment on generic and content-specific intelligent virtual agents as exhibition guides. In *Proceedings of the International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE).*

[67] Susanne Schmidt, Sven Zimmermann, Celeste Mason, and Frank Steinicke. 2022. Simulating human imprecision in temporal statements of intelligent virtual agents. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI).* 1–15.

[68] Vicki L Smith and Herbert H Clark. 1993. On the course of answering questions. *Journal of memory and language* 32, 1 (1993), 25–38.

[69] Kihyuk Sohn, Honglak Lee, and Xinchen Yan. 2015. Learning structured output representation using deep conditional generative models. *Advances in neural information processing systems* 28 (2015).

[70] Matthew Stone and Insuk Oh. 2008. Modeling facial expression of uncertainty in conversational animation. In *Proceedings of the ZiF Research Group International Workshop on Embodied Communication in Humans and Machines.* Springer, 57–76.

[71] Marc Swerts and Emiel Krahmer. 2005. Audiovisual prosody and feeling of knowing. *Journal of Memory and Language* 53, 1 (2005), 81–94.

[72] Akikazu Takeuchi and Taketo Naito. 1995. Situated facial displays: towards social interaction. In *Proceedings of the SIGCHI conference on Human factors in computing systems.* 450–455.

[73] Sarah K Tauber, John Dunlosky, Katherine A Rawson, Matthew G Rhodes, and Danielle M Sitzman. 2013. General knowledge norms: Updated and expanded from the Nelson and Narens (1980) norms. *Behavior research methods* 45 (2013), 1115–1143.

[74] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288* (2023).

[75] Anne Marthe Van Der Bles, Sander Van Der Linden, Alexandra LJ Freeman, James Mitchell, Ana B Galvao, Lisa Zaval, and David J Spiegelhalter. 2019. Communicating uncertainty about facts, numbers and science. *Royal Society open science* 6, 5 (2019), 181870.

[76] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems* 30 (2017).

[77] Thomas S Wallsten, David V Budescu, Amnon Rapoport, Rami Zwick, and Barbara Forsyth. 1986. Measuring the vague meanings of probability terms. *Journal of Experimental Psychology: General* 115, 4 (1986), 348.

[78] Isaac Wang and Jaime Ruiz. 2021. Examining the use of nonverbal communication in virtual agents. *International Journal of Human–Computer Interaction* 37, 17 (2021), 1648–1673.

[79] Elke U Weber and Denis J Hilton. 1990. Contextual effects in the interpretations of probability words: Perceived base rate and severity of events. *Journal of Experimental Psychology: Human Perception and Performance* 16, 4 (1990), 781.

[80] Katharina Weitz, Dominik Schiller, Ruben Schlagowski, Tobias Huber, and Elisabeth André. 2021. "Let me explain!": exploring the potential of virtual agents in explainable AI interaction design. *Journal on Multimodal User Interfaces* 15, 2 (2021), 87–98.

[81] Teresa Wellner. 2023. Towards a taxonomy of hedging devices in Standard German. *Regensburg Papers in Linguistics* 26 (2023).

[82] Antje Wilton. 2019. Epistemic stance markers in German and English as a lingua franca media sports interviews. *Alicante Journal of English Studies* 32 (2019), 217–243.

[83] Charlotte Wollermann, Eva Lasarcyk, Ulrich Schade, and Bernhard Schröder. 2013. Disfluencies and uncertainty perception-evidence from a human-machine scenario. In *Proceedings of Disfluency in Spontaneous Speech (DiSS).* 73–76.

[84] Yuhang Zhang, Chengrui Wang, and Weihong Deng. 2021. Relative uncertainty learning for facial expression recognition. *Advances in Neural Information Processing Systems* 34 (2021), 17616–17627.